



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

SC3021 – Data Science Fundamentals

Lab 01: Problem Formulation, Requirement Analysis, and Source Identification

Group 3, SDAC

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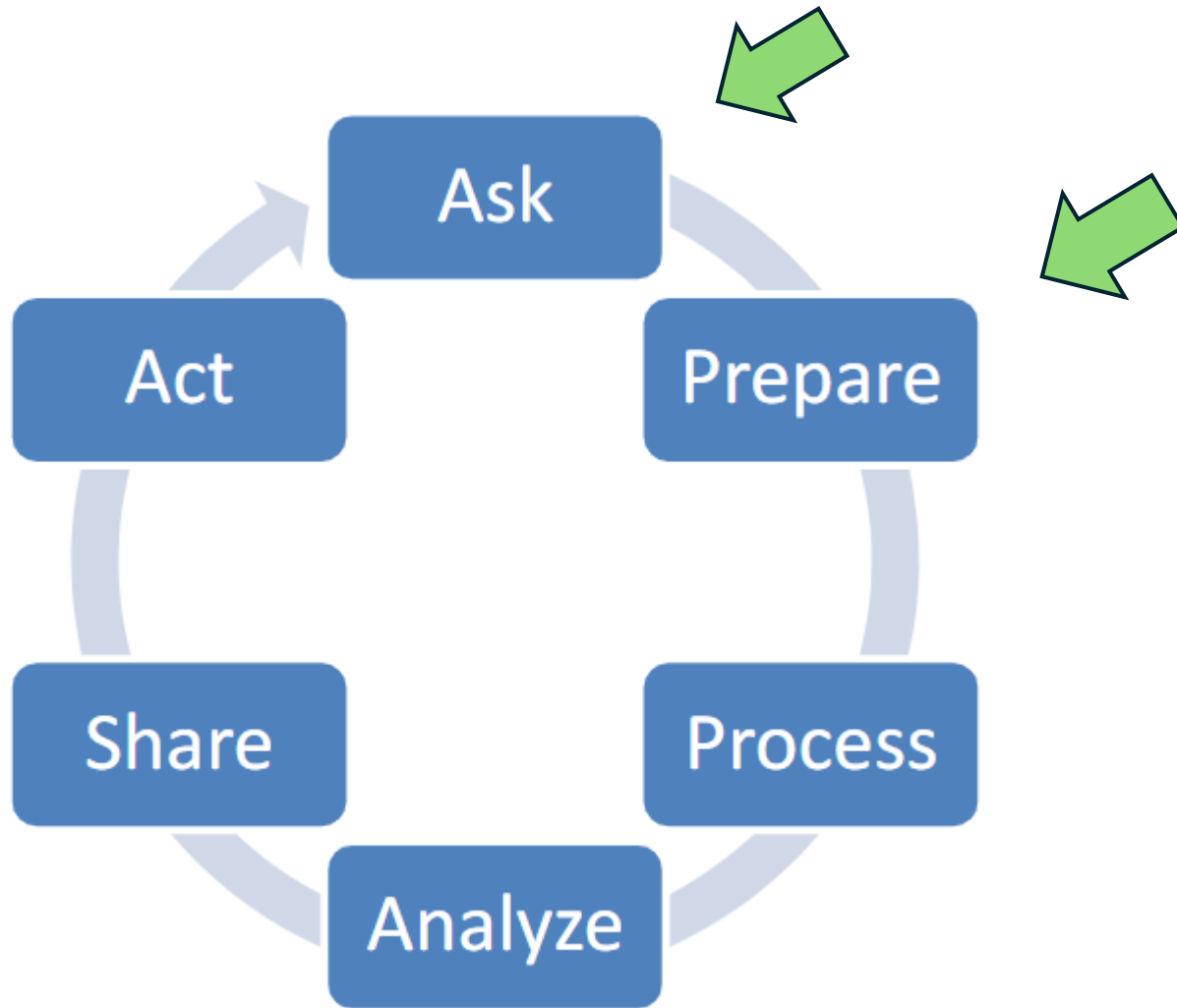
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Lab 1: Data Science Process



ASK Stage:

- Problem Formulation

PREPARE Stage:

- Requirement Analysis
- Source Identification

ASK: Problem Formulation

Background:

- Today's world of information and **technology overload**.
- Wide range of **learning methods** and **study habits**.
- **Traditional factors** that affect study performance like hours studying, class attendance, sleep patterns
- Modern learning practices like **AI-assisted learning**.
- Need for **better decision-making** in **study habits**, lifestyle factors to optimize study performance and eventually test performance



Defining the Problem



Problem to be solved: Identify key factors that influence students' ability to score >80% for their tests



Underlying Problem: Prediction of students' test performance based on a variety of factors



Goal: Identified key factors to facilitate the formulation of a robust study strategy to improve student test performance



Relevance: Useful for us as well as our friends to make better-informed decisions in pursuit of excellence

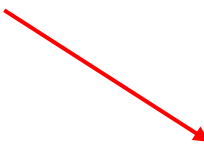
Problem

Data Analysis



“How do different learning methods and lifestyle factors influence students’ test scores, and what actionable study strategies can be recommended to improve test scores?”

Goal (Prescriptive)



Quality Metric



PREPARE (A): Requirement Analysis

Study Habits (study hours, etc.)

Academic Performance (test Scores)

Must Capture
Student-Level Data

Lifestyle Factors (sleep patterns, etc.)

AI Assistance (frequency, etc.)

Data Requirements

Diverse group of students (wide range of study habits, various lifestyles, different levels of AI usage, etc.)

- To enable meaningful comparisons across learning approaches

One aggregated record per student is sufficient

- Focus is on overall academic performance, not short-term behavior

Fine-grained temporal data is not required

- No need for continuous tracking

PREPARE (B): Source Identification

Three different datasets (Kaggle):

- Dataset on **Traditional Learning Behaviour.**
- Dataset on **Student's Lifestyle.**
- Dataset on **Usage of AI in Studies.**

DS1. Traditional Learning Behavior

- Mainly gives info about students' Learning behavior.
- Important features include:
 - Study hours
 - Class attendance
 - Sleep hours
 - Sleep quality
 - Internet access
- Obtained from
 - <https://www.kaggle.com/competitions/playground-series-s6e1/data>

ds1_study-habit

id	age	gender	course	study_hours	class_attendance	internet_access	sleep_hours	sleep_quality	study_method	facility_rating	exam_difficulty	exam_score
0	21	female	b.sc	7.91	98.8	no	4.9	average	online videos	low	easy	78.3
1	18	other	diploma	4.95	94.8	yes	4.7	poor	self-study	medium	moderate	46.7
2	20	female	b.sc	4.68	92.6	yes	5.8	poor	coaching	high	moderate	99.0
3	19	male	b.sc	2.0	49.5	yes	8.3	average	group study	high	moderate	63.9
4	23	male	bca	7.65	86.9	yes	9.6	good	self-study	high	easy	100.0
5	24	male	b.com	5.04	85.1	yes	9.4	average	online videos	medium	moderate	70.1
6	20	male	b.sc	4.28	87.0	no	9.1	average	mixed	high	moderate	63.4
7	22	female	ba	4.19	44.9	yes	8.8	good	self-study	high	hard	76.8
8	22	other	b.com	1.06	98.3	yes	5.0	poor	mixed	low	moderate	46.7
9	18	male	bba	3.44	80.9	yes	6.2	good	group study	medium	easy	58.2
10	20	other	bba	7.24	51.6	yes	7.4	good	self-study	high	easy	85.9
11	19	female	b.tech	6.85	43.2	yes	8.5	poor	self-study	high	moderate	74.5
12	24	other	ba	7.4	92.4	yes	8.2	average	coaching	high	moderate	100.0
13	19	other	bba	6.99	90.5	yes	6.4	good	mixed	high	moderate	93.5
14	21	male	b.com	2.91	78.4	yes	6.7	average	online videos	low	easy	51.6
15	23	male	b.sc	7.47	64.8	yes	9.3	poor	self-study	low	moderate	76.8
16	23	male	b.tech	4.56	63.9	yes	8.4	average	mixed	high	moderate	58.6
17	21	male	bba	3.8	53.3	yes	4.7	good	coaching	high	moderate	67.4
18	17	female	b.tech	0.08	47.7	yes	9.0	average	coaching	low	moderate	32.2
19	17	male	b.tech	4.39	97.9	yes	9.1	good	group study	high	moderate	94.3
20	22	female	b.sc	6.23	92.3	yes	4.8	average	mixed	medium	moderate	77.4
21	22	female	bca	0.12	78.6	yes	4.1	poor	mixed	low	moderate	34.5
22	19	female	b.com	7.91	87.2	yes	8.2	good	online videos	high	easy	92.4
23	17	other	b.com	6.75	44.2	yes	9.9	good	group study	low	moderate	80.8
24	20	male	diploma	4.01	54.5	yes	5.4	good	group study	low	moderate	49.5
25	22	other	b.com	7.59	84.0	yes	5.0	average	self-study	high	moderate	76.2
26	21	male	ba	7.66	97.3	yes	9.0	good	online videos	high	easy	91.2
27	17	other	bca	3.73	41.2	yes	8.6	good	self-study	high	moderate	62.3
28	20	male	b.com	0.49	51.2	yes	7.9	average	self-study	medium	easy	31.5
29	18	male	b.tech	5.08	77.5	yes	7.4	average	mixed	high	moderate	77.4
30	24	female	b.com	2.67	63.6	yes	6.9	good	group study	high	moderate	69.1
31	19	female	b.com	3.99	53.0	no	7.7	good	self-study	high	moderate	57.3
32	23	male	b.com	0.4	46.9	yes	4.1	average	online videos	medium	easy	42.9
33	17	female	bca	2.64	72.3	no	8.5	good	self-study	high	hard	53.1

... up to 630,000 rows

DS2. Students' Lifestyles

- Mainly gives info about students' lifestyles outside academics
- Important features include:
 - Social media hours
 - Netflix hours
 - Extracurricular participation
 - Diet Quality
 - Parents' education level
- Obtained from
 - <https://www.kaggle.com/datasets/jaantanaath/student-habits-vs-academic-performance>

ds2_lifestyle												
student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_time_job	attendance_percentage	sleep_hours	diet_quality	exercise_frequency	parental_education_level	internet_quality
S1000	23	Female	0.0	1.2	1.1	No	85.0	8.0	Fair	6	Master	Average
S1001	20	Female	6.9	2.8	2.3	No	97.3	4.6	Good	6	High School	Average
S1002	21	Male	1.4	3.1	1.3	No	94.8	8.0	Poor	1	High School	Poor
S1003	23	Female	1.0	3.9	1.0	No	71.0	9.2	Poor	4	Master	Good
S1004	19	Female	5.0	4.4	0.5	No	90.9	4.9	Fair	3	Master	Good
S1005	24	Male	7.2	1.3	0.0	No	82.9	7.4	Fair	1	Master	Average
S1006	21	Female	5.6	1.5	1.4	Yes	85.8	6.5	Good	2	Master	Poor
S1007	21	Female	4.3	1.0	2.0	Yes	77.7	4.6	Fair	0	Bachelor	Average
S1008	23	Female	4.4	2.2	1.7	No	100.0	7.1	Good	3	Bachelor	Good
S1009	18	Female	4.8	3.1	1.3	No	95.4	7.5	Good	5	Bachelor	Good
S1010	19	Female	4.6	3.7	0.8	No	77.6	5.8	Fair	1	None	Good
S1011	23	Male	3.9	2.4	2.5	No	71.7	7.9	Fair	2	Bachelor	Average
S1012	19	Female	3.7	2.1	0.4	Yes	81.1	4.5	Fair	1	Bachelor	Good
S1013	19	Female	3.4	2.7	2.7	No	89.3	4.7	Fair	4	Bachelor	Good
S1014	24	Male	2.4	1.5	0.7	No	87.4	6.7	Poor	6	Bachelor	Average
S1015	21	Male	3.1	5.0	1.0	No	97.5	6.5	Good	6	High School	Average
S1016	20	Male	1.0	0.6	0.2	No	92.9	5.6	Poor	3	High School	Poor
S1017	24	Female	3.4	2.7	1.2	No	94.7	7.5	Poor	0	High School	Average
S1018	24	Other	2.0	4.9	2.9	Yes	88.3	7.1	Good	2	High School	Good
S1019	19	Male	1.8	2.5	2.4	No	71.1	7.5	Fair	1	High School	Average
S1020	22	Female	3.8	2.3	2.9	No	83.0	6.4	Good	3	Master	Good
S1021	21	Male	5.6	2.1	2.4	No	95.6	7.2	Fair	1	High School	Good
S1022	18	Other	4.9	2.3	0.6	No	84.5	6.0	Fair	3	High School	Average
S1023	24	Female	1.1	4.1	1.4	Yes	90.0	9.0	Fair	6	Bachelor	Good
S1024	20	Female	2.0	0.0	0.9	Yes	81.8	5.5	Fair	4	Bachelor	Average
S1025	22	Male	4.9	4.3	3.3	No	74.7	9.0	Fair	1	High School	Average
S1026	22	Male	2.0	0.8	0.5	No	83.8	6.5	Good	4	Bachelor	Poor
S1027	18	Male	3.2	2.2	2.8	Yes	88.1	4.8	Fair	5	Bachelor	Average
S1028	24	Male	4.3	2.0	2.8	Yes	78.4	8.1	Good	5	High School	Poor
S1029	20	Female	2.0	3.2	3.8	Yes	82.6	6.7	Poor	6	High School	Poor
S1030	21	Female	3.7	0.6	1.3	No	75.6	7.5	Fair	2	Master	Average
S1031	17	Other	1.5	3.1	2.6	Yes	96.2	8.0	Fair	4	Bachelor	Average
S1032	20	Other	2.6	4.9	4.3	No	82.1	6.9	Good	3	Bachelor	Good

... up to 2,000 rows

DS3. Usage of AI in Studies

- Mainly gives info about students' usage of AI in their studies
- Important features include:
 - AI usage
 - Name of AI tool used
 - Percentage of AI generated content
 - AI ethics score
- Obtained from
 - <https://www.kaggle.com/datasets/ankushnarwade/ai-impact-on-student-performance>

student_id	age	gender	grade_level	study_hours_per_day	uses_ai	ai_usage_time_minutes	ai_tools_used	ai_usage_purpose	ai_dependency_score	ai_generated_content_percentage	ai_prompts_per_week	ai_ethics_score
1	20	Female	1st Year	2.5	1	170	None	Exam Prep	10	29	56	5
2	17	Male	12th	3.4	1	123	None	Notes	4	14	65	1
3	24	Male	3rd Year	0.8	0	35	Copilot	Doubt Solving	8	82	8	5
4	21	Female	12th	4.4	0	45	ChatGPT+Gemini	Notes	3	74	106	10
5	18	Other	3rd Year	3.5	1	21	ChatGPT+Gemini	Coding	2	83	7	10
6	20	Male	1st Year	5.1	1	34	Copilot	Coding	10	24	95	6
7	23	Female	1st Year	2.0	1	112	ChatGPT+Gemini	None	8	37	57	5
8	16	Male	3rd Year	5.5	0	174	ChatGPT	Homework	4	42	70	6
9	20	Female	10th	6.0	1	32	Copilot	None	9	29	37	9
10	24	Female	1st Year	5.6	0	140	None	Doubt Solving	3	39	19	7
11	24	Female	10th	4.8	1	55	Copilot	Homework	9	58	46	5
12	21	Male	10th	5.0	0	66	Copilot	Homework	10	22	78	10
13	18	Female	3rd Year	3.7	1	12	ChatGPT+Gemini	Coding	6	43	102	9
14	17	Female	10th	1.5	1	49	Copilot	Coding	5	80	27	5
15	21	Male	10th	4.2	1	27	Gemini	None	2	0	94	5
16	21	Male	11th	4.4	0	103	Gemini	Doubt Solving	6	77	59	6
17	16	Male	12th	5.9	1	97	ChatGPT	Coding	4	81	21	8
18	19	Male	10th	3.5	1	95	Copilot	Doubt Solving	5	40	103	1
19	18	Female	12th	2.0	0	86	Copilot	Notes	6	61	112	9
20	15	Male	10th	4.3	1	87	None	None	7	59	50	8
21	21	Male	12th	5.9	1	151	ChatGPT+Gemini	Exam Prep	4	79	23	7
22	19	Female	1st Year	1.5	1	124	None	None	8	9	97	5
23	15	Male	1st Year	2.3	0	80	Claude	Notes	5	72	72	1
24	18	Female	12th	3.5	1	6	None	Coding	7	77	60	7
25	14	Male	10th	1.4	1	73	Claude	Exam Prep	6	65	1	9
26	23	Female	11th	0.6	1	82	Copilot	Exam Prep	5	56	108	3
27	19	Female	12th	2.4	0	23	ChatGPT	None	7	41	88	8
28	22	Female	2nd Year	5.3	1	136	Gemini	None	4	53	75	3
29	14	Male	11th	0.7	1	113	Copilot	Homework	7	84	54	3
30	24	Female	2nd Year	2.9	0	14	Copilot	Exam Prep	3	41	3	7
31	24	Male	12th	4.7	1	112	ChatGPT	Doubt Solving	8	65	114	6
32	23	Male	10th	5.7	1	121	ChatGPT	Coding	10	69	103	5
33	16	Male	2nd Year	5.9	0	85	Claude	Coding	4	10	57	7
34	20	Female	12th	2.8	1	144	Copilot	Homework	4	66	12	9

... up to 8,000 rows